Sim2Real Hackathon
The Postman Problem

Organised by: Robotech NITK | **Venue:** Hybrid

1. Introduction

The **Sim2Real Robotics Hackathon** is a two-stage competitive challenge designed to bridge the gap between simulation algorithms and physical hardware deployment.

While ideal simulations assume noise-free sensing and perfect actuation, real-world robots suffer from wheel slippage, encoder noise, sensor latency, and physical tolerances. This competition evaluates whether an autonomy stack developed in simulation is architecturally robust enough to transition seamlessly to bare-metal embedded hardware.

The core objective across both rounds is autonomous point-to-point delivery:

A differential-drive mobile robot must solve an online dynamic Traveling Salesperson Problem (TSP) - autonomously discovering, sequencing, and navigating to a field of targets with minimum path distance and zero collisions, fulfilling mandatory dwell stops at each target, and returning safely to the home base.

- **Round 1 (Virtual Postman):** Hosted in a Gazebo Harmonic and ROS 2 Jazzy environment. Contestants program an autonomous differential-drive robot across three progressive tiers involving noisy encoder/IMU sensor fusion, overhead camera perception, procedural target spawning, and obstacle avoidance.

- **Round 2 (Embedded Hardware):** Shortlisted finalist teams deploy their control, estimation, and path-planning algorithms directly onto physical differential-drive hardware powered by an ESP32 microcontroller using bare-metal C/C++.

Round 1 is a qualifier. Only teams satisfying minimum performance benchmarks on the automated evaluation harness will advance to Round 2.

2. Eligibility and Team Structure

1. **Team Size:** Teams must comprise **2 to 4** members.
2. **Single-Team Invariant:** Each participant may belong to only one team across the entire competition. Multiple team registrations by a single individual will result in immediate disqualification of all associated teams.
3. **Academic Status:** All participants must be actively enrolled undergraduate or postgraduate students at a recognized educational institution. Valid student identification cards must be presented upon request.
4. **Inter-College Teams:** Inter-institutional teams are permitted.
5. **Team Lead Designation:** Each team must designate a single Team Lead during registration. The Team Lead acts as the official liaison for technical updates, submissions, and official communications.
6. **Roster Freeze:** Team composition is locked upon submission of the Round 1 qualifying solution. No member substitutions, additions, or transfers are permitted between Round 1 and Round 2 without prior formal approval from the organizing committee.

3. Schedule

Event	Date and Time
Registration Opens	7th October, 2026
Registration Closes	14th October, 2026
Round 1 Submission Deadline	17th October, 2026
Shortlist Announcement	20th October, 2026
Round 2 Begins (Hardware Kits Issued)	24th October, 2026
Round 2 Build Deadline	25th October, 2026
Results and Prize Distribution	30th October, 2026

* This schedule is subject to change. Please check the website for the latest updates.

4. Round 1: The Virtual Postman (ROS 2 and Gazebo)

4.1 Objective

Teams must configure an autonomous differential-drive robot to navigate an arena, visit all designated target waypoints in an optimal sequence, and return to the start base. Two additional dynamic waypoints will spawn mid-run and must be detected and incorporated into the route.

Round 1 consists of three progressive evaluation tiers:

- **Tier 1:** Target coordinates are published directly via ROS 2 topics.
- **Tier 2:** Coordinate topics are withheld; targets must be detected visually using an overhead camera.
- **Tier 3:** Cluttered environment; targets and static obstacle walls must both be detected visually, requiring collision-free path planning.

4.2 Provided Environment & Stack

Teams are provided with a complete ROS 2 workspace containing two packages:

postman_pkg (Simulation Platform - Do Not Modify):

- Complete robot description (URDF) with realistic wheel friction, mass parameters, and sensor plugins.
- 3.0 m x 3.0 m digital-twin arena environments (clean and obstacle-laden) for Gazebo Harmonic.
- Procedural waypoint generator node with seeded placement.
- Automated grading script (evaluator.py) to verify scores locally.

postman_solution (Participant Workspace - Work Here):

- Starter template nodes (tier1_node.py, tier2_node.py, tier3_node.py).
- Submission launch file (solution.launch.py).
- Teams must write all their navigation, perception, and control code inside this package.

* Complete breakdown of the different tasks is provided in the **README.md** file.

4.3 Waypoint Definition

- **Coordinates:** All spatial coordinates are defined in meters within the 2D arena frame (**odom**).

- **Start Zone:** The robot begins at coordinates (0.0, 0.0) with a heading of 0.0 radians (facing +X).
- **Target Layout:**
  -> 5 initial static waypoints (green visual disks, radius 0.15 m).
  -> 2 dynamic waypoints spawned during the run (at T+5.0 s for Tier 1; T+7.0 s for Tiers 2 and 3), bringing the total to 7 targets.
- **Collection Criteria:** A waypoint is officially registered as collected when:
  1. The robot's center comes within 0.20 m of the target center.
  2. The robot comes to a continuous stop (|v| <= 0.03 m/s) at the target for at least 2.00 seconds.
- **Home Return:** The run officially concludes when the robot returns to within 0.20 m of (0.0, 0.0) and dwells for at least 2.00 seconds.

4.4 Tasks and Scoring (100 points)

Task	Description	Points
State Estimation and TF	Fuse wheel encoder velocities and IMU data to estimate robot pose. Publish nav_msgs/Odometry on /odom and broadcast the dynamic odom to base_link transform using TF2 for RViz tracking.	10
Route Planning and Dwell Navigation	Autonomously sequence and navigate the 5 initial waypoints publishing to /cmd_vel. Fulfill the 2.00-second continuous stop dwell requirement (|v| <= 0.03 m/s) at each target and return safely to the start base.	20
Dynamic Target Insertion	Handle the 2 dynamic waypoints spawned mid-run. Detect their release, compute an optimal insertion into the remaining unvisited route, and navigate to all 7 total waypoints without reordering completed targets.	30
Vision and Obstacle Avoidance (Tiers 2 & 3)	Detect green target disks visually from /overhead_camera/image_raw when coordinate topics are withheld (Tier 2), and plan collision-free paths around the 4 interior obstacle walls and perimeter boundaries (Tier 3).	40

4.5 Rules

1. The Dwell Rule (2.00 Seconds):
At every waypoint, the robot must come to a continuous stop within the 0.20 m capture radius and remain stationary for at least 2.00 seconds before proceeding.

* Stationary is defined as measured linear velocity |v| <= 0.03 m/s.
* A robot that rolls or cuts through a waypoint without fulfilling the continuous 2.00-second dwell will not receive credit for that target.

2. Dynamic Waypoints:

Two additional dynamic waypoints are spawned mid-run (at T+5.0 s for Tier 1; T+7.0 s for Tiers 2 and 3) after the robot initiates movement. The autonomy stack must detect these targets and dynamically insert them into the remaining navigation queue.

3. Control & Sensor Restrictions:

* **Prohibited Topics:** Subscribing to **/evaluator** topics or any internal Gazebo ground-truth poses (**/gz**) is strictly forbidden and results in immediate disqualification. State estimation must rely strictly on **/joint_states**, **/imu**, and **/overhead_camera**.

* **Prohibited Stacks:** Pre-built autonomous navigation suites (e.g., Nav2) are not permitted. Teams must write their own state estimation, path planning, and trajectory-tracking controllers. Standard ROS 2 utility libraries (tf2, geometry_msgs, OpenCV, cv_bridge) are fully permitted.

* **Hardcoding Restriction:** Hardcoding coordinates is prohibited. Submissions will be evaluated against unseen random seeds.

4. Run Time Limit

Each run has a maximum time limit of 10 minutes (simulation time). If a run exceeds 10 minutes, it is terminated and scored based on progress achieved up to that point.

Official Environment:

Operating System: Ubuntu 24.04 LTS
Middleware: ROS 2 Jazzy Jalisco
Simulator: Gazebo Harmonic

4.6 Submission

Teams must submit the following before the specified deadline:

1. **Workspace Archive:** A single ZIP archive or repository link containing only the **src/** directory and **README.md**. Build artifacts (build/, install/, log/`) must be excluded.

2. **Demonstration Video:** A screen capture demonstrating a successful run in Gazebo and RViz.

3. **Penalty for Late Submission:** Submissions delayed past the deadline will incur a penalty of 10 points per 20 - minute interval, up to a maximum delay of 90 minutes. Submissions past 60 minutes will not be accepted.

4.7 Round 1 Evaluation

Evaluations are conducted by the judging committee on a standardized testing machine using an unseen procedural seed:

1. **Launch:** The organizer executes **sim.launch.py** with the evaluation seed and runs the team's **solution.launch.py**.

2. **Scoring:** The run is recorded and officially graded using the automated **evaluator.py** script, which tracks waypoints collected, dwell compliance, collisions, home return, odometry distance, and elapsed simulation time.

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5. Round 2: Embedded Hardware

5.1 Objective

Build a physical differential drive robot, read raw encoder and IMU data on the ESP32, implement closed-loop control, and execute a TSP route using dead reckoning alone.

5.2 Hardware Kit

Each team is issued:

- ESP32 development board
- Two N20 motors with Hall-effect encoders
- MPU6050 IMU
- BTS7960 motor driver
- USB Port
- 3S LiPo battery
- 3D-printed chassis
- [Wheels, caster, fasteners, wiring, breadboard / perfboard, connectors]

Teams may bring their own tools, soldering equipment, laptops and passive components (resistors, capacitors, connectors, wire, heat shrink). Additional sensors, microcontrollers, motors, drivers or batteries are not permitted. Kit components must be returned undamaged where specified by the organisers.

5.3 Arena

- Arena size: **240cm x 240cm**
- Surface: Matte frontlit vinyl flex banner, taped flat to the floor
- Origin: marked on the arena floor. The robot starts with its wheel axle centre on the origin, facing the +x axis.
- Coordinate system: x runs along horizontal, y along vertical, units in centimetres. Positive θ is counter-clockwise.
- Waypoints are marked by circles of **10-15 cm radius**, centred on the given coordinate.

5.4 Tasks and Scoring (100 points)

Task	Description	Points
Sensor Fusion and Odometry	Read encoder ticks through hardware interrupts and IMU yaw from the MPU6050. Combine them to produce real-time x, y and θ. The estimate must be shown live on serial output or a display.	25
Point-to-Point PID Control	Demonstrate closed-loop control. The robot must rotate to a commanded heading and drive a commanded distance with minimal overshoot.	25
Hardware Cable Management	Judged on wiring neatness, secure and safe component placement, and structural assembly.	10
Route Execution	Five coordinates are entered, the code is compiled and uploaded, and the robot is placed at the origin. It must compute the optimal visiting order on its own and visit all waypoints	40

5.5 Task Details

Task 1: Odometry. Encoder reading must use interrupts, not polling in the main loop. Judges will ask the team to push the robot or rotate it by hand and will check that the reported pose changes plausibly. Fused heading must be demonstrated as using the IMU.

Task 2: PID demonstration. Judges issue two commands in sequence on the arena: a rotation to a specified angle (for example 90°) and a straight drive of a specified distance (for example 100 cm). Scoring is based on final error against the commanded values and on overshoot and oscillation. The commands are given at evaluation time and may be changed per team.

Task 3: Cable management. Judged on a checklist: no loose or hanging wires, wires clear of wheels, battery secured and accessible, strain relief on connectors, no exposed live conductors, logical component layout, and a power switch or easily accessible disconnect. A robot that fails the safety items in Section 5.8 is not scored on this task until corrected.

Task 4: Route execution.

- Coordinates are released **15 minutes before** each team's run slot and are entered by the team into the source code (or an equivalent input method approved by the judges). Teams must not use pre-computed routes: the optimal order must be computed on the ESP32.
- The route is an open path starting at the origin. The robot stops at each waypoint and pauses for at least 2 seconds so that judges can record the position, then continues. The robot must indicate a stop clearly. After visiting the waypoints, the robot is expected to return to the start zone.
- Navigation must rely on onboard sensing and dead reckoning only. No external positioning, tracking, line following, floor markers or remote control.
- Between code upload and the start of the run, the robot may not be touched or reprogrammed. Once the run starts, it must be fully autonomous.

5.6 The 10 cm Rule (Handling Drift)

- **Hit:** after the robot stops at a waypoint, any part of its chassis touches or lies inside the 10 cm circle. The full points for that waypoint are awarded.
- **Miss:** the robot stops with no part of the chassis touching the circle. Points for that waypoint are not awarded.
- **Recovery:** after a miss, the team may pick up the robot and place it on the missed waypoint, then let it continue. The robot's internal position estimate is then to be reset by the team's code to that waypoint's coordinate, in whatever way the team has implemented it (for example, a button press or serial input). Time spent handling the robot counts against the run time.
- A waypoint is judged at the moment the robot comes to rest, as decided by the referee in the arena. Referee decisions on hit and miss are final.

Route Execution points are split equally across the five waypoints. The final score is described in Section 6.

5.7 Run Rules

1. Each team has **3 attempts**, with a 10-minute run window each. The better of the two is retained for Task 4.
2. Each run is timed from the signal to start until the robot comes to rest at the last waypoint, or until the window ends.
3. Robot hand placement, other than the recovery procedure in 5.6, is not allowed during a run. Every unauthorised touch is a 5-point penalty.
4. The robot must not leave the arena. A robot that leaves the arena is stopped, and the run ends or the team recovers as in 5.6.

5. Deliberate contact with arena walls to correct position is not permitted unless the organisers state that walls are part of the arena.

5.8 Safety Requirements

LiPo batteries are fire hazards. The following are mandatory:

1. The battery must be physically secured to the chassis and protected from impact and puncture.
2. Charging is only allowed at the designated charging area, under supervision, with a LiPo-rated charger. Charging unattended, or in the build area, is prohibited.
3. A battery that is swollen, punctured or damaged must be reported immediately and may not be used.
4. No bare conductors or exposed terminals. Wiring must be insulated and rated for the load.
5. The robot must have a means of cutting battery power quickly (a switch or an accessible connector).
6. Organisers and volunteers may stop any run or remove any robot from the arena for safety reasons.

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6. Scoring Summary and Tie-Breaks

Round 1 is out of 100, and Round 2 is out of 100. The final ranking is based on a weighted combination: 30% Round 1 and 70% Round 2.

Round 1 results are used to select teams for Round 2.

Ties are broken in this order:

1. Higher Route Execution score (Round 2, Task 4).
2. Fewer recoveries / hand placements during the route run.
3. Shorter total run time on the best attempt.
4. Higher Round 1 score.
5. Judges' decision.

7. General Rules and Conduct

1. **Original work.** All code must be written by the team during the event. Open-source libraries (for example sensor drivers, Arduino core, standard ROS 2 packages) are allowed and must be acknowledged in the README. Copying another team's code, or using code from a previous implementation of this specific problem, is prohibited.
2. **AI tools.** Allowed, but teams are responsible for being able to explain every part of their submission.
3. **Viva.** Judges may ask any team member to explain their code, algorithm or wiring. Inability to explain submitted work may reduce scores.
4. **Fair play.** Interference with another team's robot, equipment or network, and any attempt to tamper with the evaluation setup, results in disqualification.
5. **Conduct.** Participants must follow venue rules and instructions from organisers. Harassment or abusive behaviour results in removal from the event.
6. **Equipment.** Teams are responsible for the kit issued to them. Damage caused by negligence, including by incorrect wiring or polarity, is the team's responsibility.
7. **Changes.** Organisers may amend the rules and schedule if needed. Changes will be announced and apply to all teams.

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8. Disqualification

A team may be disqualified for:

- Plagiarism or sharing code between teams.
- Unsafe handling of batteries after a warning.
- Tampering with the arena, evaluation system or another team's equipment.
- Violation of any rule that the judges consider to be a serious breach.

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9. Disputes and Judging

1. Judges' and referees' decisions on the arena are final.
2. A team that wishes to raise an objection must do so through the team lead, in writing, within 15 minutes of the result being posted for the run in question.
3. Objections must refer to a specific rule in this document. The decision of the organising committee on any objection is final.

10. Contact

Role	Name	Contact
Event Head	Vivek Jain	+91 81849 89849
Event Head	Aditya Kumar Sahu	+91 76762 02418